SOURCE SECTION 1

Simulated Teaching Hospital / Spine Surgery

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Lumbar Disc Herniation: Complete Medical Record and Imaging

Left-sided L4/5 disc extrusion with progressive left L5 nerve root motor impairment

Case information

Field	Details
Patient	Li Zhiqiang (fictional), male, 46 years old
Medical record number	SIM-LDH-20260903
Outpatient visit / Discharge	Outpatient consultation: 2026-09-03 09:00; discharge: 2026-09-07 10:00; the subsequent inpatient admission processing time has not been added to the record
Surgery	2026-09-04: Left-sided L4/5 microsurgical discectomy and nerve root decompression
Review status	v1.1 · Data synchronized on 2026-09-14; pending review by clinical and imaging experts
Imaging coverage	Lumbar spine X-rays: anteroposterior, lateral, and flexion–extension views; CT: axial, coronal, and sagittal views; MRI: T1- and T2-weighted sagittal views and T2-weighted axial views

Instructions for Use

This document and the accompanying PACS contain a fictional case for teaching. The history, laboratory results, examination reports, surgery, and clinical course are stipulated elements of the case. The images were generated by AI, and their morphology may not fully match the report specifications. Imaging experts must independently verify them; image accuracy must not be inferred from the reports.

Identified issue: The three lumbar X-ray series-lateral, flexion, and extension-contain too many lumbar-like vertebral bodies and do not meet this case's specification of five lumbar vertebrae. One targeted correction still failed to reliably resolve this issue. The original images without misleading numbering have been retained, and each series has been marked as unacceptable imaging, for interface demonstration only. The T1-weighted sagittal images have been corrected using the T2-weighted images as a reference but remain pending expert review. The CT and MRI axial series have each been expanded to 16 representative levels from the inferior margin of T12 to S2, with four images in each of the remaining planes; these are not clinical thin-slice continuous scans. The sagittal localizer in the upper left corner and the reference line that moves with the slices are manually configured approximate correspondences, not DICOM spatial registration. Sacral morphology and vertebral body/intervertebral disc boundaries are synthetic illustrations. The localizer also shows an L3/4 disc bulge; experts must verify its consistency with the axial images and reports. Each plane was synthesized independently, so three-dimensional anatomical consistency and continuity between slices cannot be guaranteed. There are no real HU values, pixel spacing, DICOM orientation matrices, or examination dose data. The PACS only simulates series selection, scrolling, zooming, panning, grayscale inversion, display brightness/contrast adjustment, and four-window comparison; it does not provide diagnostic-grade measurements or true MPR.

The PACS allows browsing of 52 representative images (10 asset groups), not a complete continuous cross-sectional imaging examination. Flexion–extension X-rays are for image demonstration only; true segmental range of motion must not be calculated from independently synthesized images.

This version is intended for expert review and interactive teaching demonstrations. It has no real hospital seal, signature, DICOM acquisition information, or formal diagnostic validity.

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SOURCE SECTION 2

Simulated Teaching Hospital / Department of Spine Surgery

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Registration Information and Synchronization Notes

Registration information

Field	Details
Name	Li Zhiqiang
Sex	Male
Age	46 years
Marital status	Married
Occupation	Warehouse management personnel
Ethnicity	Han Chinese (teaching registration setting)
Place of origin	Chengdu, Sichuan Province (teaching registration setting)
Address	Apartment 502, Building 3, No. 26 Anhe Road, Wuhou District, Chengdu (fictional address for teaching)
History provided by	The patient and his spouse
Visit registration status	General orthopedic outpatient clinic; registration information complete; inpatient admission procedures not yet completed

Source task: 1.1 Lumbar disc herniation case. The medical master draft remains v1.1, pending review by clinical and imaging experts. This synchronization adds registration information, the outpatient consultation scenario, and a fixed interview background; it does not add an inpatient admission processing time or change facts concerning surgery, muscle strength, or laboratory tests.

For the 19-item medication catalog and notes on its sources and dosage review, see the accompanying complete materials page and downloadable attachments; inclusion in the medication database does not mean that the patient actually used the medication. The original case record is retained separately.

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Initial Outpatient Assessment and Complete History

General Information

Male, 46 years old, married, warehouse management worker. History provided by the patient and his spouse; reliable. Department: general orthopedic outpatient clinic; not yet admitted at the time of assessment. No real identity, address, telephone number, or hospital code is included. Name: Li Zhiqiang (fictional for teaching purposes).

Clarification of Outpatient and Inpatient Stages

The encounter at 09:00 on 2026-09-03 was in the general orthopedic outpatient clinic; admission had not yet been arranged. Subsequent hospitalization and surgery are separate teaching stages. The admission time in the original case record is retained in the source attachment and is no longer taken to mean that the patient was already hospitalized at the outpatient encounter. Original examination dates and simulated performance times following requests during the exercise are recorded separately; no additional medical interventions are introduced on this basis.

Chief Complaint

Low back pain with radiating left lower limb pain for 8 weeks, worsening with left foot dorsiflexion weakness for 2 days.

History of Present Illness

Lower back pain began on July 9, 2026, after bending over to move boxes. The next day, radiating pain and numbness developed in the left buttock, extending along the posterolateral left thigh to the lateral calf, dorsum of the foot, and first interdigital web space. Symptoms worsened with coughing, sneezing, prolonged sitting, and bending, and improved when lying on his side with the hips and knees flexed. Initially, low back pain was NRS 4/10 and leg pain 6/10; currently, leg pain is 8/10 and low back pain 4/10. He needs to stop after walking approximately 100 meters and wakes at night because of pain. Bilateral intermittent claudication is not the predominant symptom pattern.

Course of Conservative Treatment

After symptom onset, approximately 8 weeks of nonoperative outpatient treatment were provided for suspected lumbar disc herniation, including a short course of celecoxib 200 mg once daily, acetaminophen as needed, rehabilitation exercises that did not provoke leg pain, and modification of lifting and prolonged sitting activities. NSAIDs were not used continuously for the full 8 weeks. Pain recurred after temporary relief, and work and sleep remained markedly limited. No lumbar injections or previous surgery. The outpatient record dated 2026-08-28 documented left great toe dorsiflexion strength of 4/5 and strength of 5/5 in the other major muscle groups; imaging was performed to assess bony structures and stability.

Recent Progression

From September 1, he had difficulty lifting the left foot, with the toes tending to catch on the ground. This worsened on September 2, and he could not walk on the left heel alone. At the outpatient assessment at 09:00 on September 3, left ankle dorsiflexion strength was 3/5 and great toe dorsiflexion strength was 3/5. Because of progressive motor decline, MRI was prioritized and completed that morning, along with a spine surgery assessment; early decompression surgery was performed the following day. Neurological assessment was not delayed to await a longer course of conservative treatment or completion of X-rays/CT.

Red-Flag Symptoms and Relevant Negative History

No urinary retention, urinary incontinence, fecal incontinence, perineal numbness, or new-onset sexual dysfunction; no simultaneous weakness of both lower limbs. No fever, chills, progressively worsening nocturnal pain at rest, or unexplained weight loss; no history of malignancy, immunosuppression, intravenous drug abuse, or recent infection. No major trauma or long-term steroid use. Appetite is fair, sleep has recently been affected by pain, and bowel and bladder function are normal.

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Past Medical History

Denies a history of hypertension, diabetes mellitus, coronary artery disease, stroke, chronic kidney disease, or gastrointestinal ulceration or bleeding. No history of spinal surgery, anesthesia-related adverse events, bleeding disorders, or venous thrombosis. No long-term anticoagulant or antiplatelet use.

Medication and Allergy History

Intermittent celecoxib use at 200 mg/day over the past week; last taken on 2026-09-02. Occasional acetaminophen use. Denies known allergies to medications, latex, or contrast agents. All imaging is specified as noncontrast, with no contrast exposure.

Personal, Marital, Reproductive, and Family History

Smoking history of approximately 15 pack-years, averaging 10 cigarettes/day in recent years; occasional alcohol use. Alternates between prolonged sitting and standing at work and previously lifted heavy objects frequently. Married with one daughter; good family support. No family history of spinal deformity or hereditary neuromuscular disease.

General Physical Examination

T 36.6°C, P 78 beats/min, R 18 breaths/min, BP 128/78 mmHg, SpO_2 98%. Height 176 cm, weight 78 kg, BMI 25.2 kg/m^2. Alert, with a pained facial expression. No obvious abnormalities on cardiac, pulmonary, or abdominal examination. No lower-extremity swelling; dorsalis pedis pulses are symmetrical.

Spinal and Neurological Examination

Mild rightward trunk shift while standing and limited lumbar forward flexion. Left paraspinal tenderness at L4/5, with no obvious bony pain on percussion. Left straight-leg raising at 35° elicits typical ipsilateral radiating pain; the ankle dorsiflexion augmentation test is positive. Right straight-leg raising at 70° produces no ipsilateral leg pain; the crossed straight-leg raise test is negative. Reduced pinprick sensation over the lateral left lower leg, dorsum of the foot, and first interdigital web space. Patellar and Achilles reflexes are 2+ bilaterally; pathological reflexes are negative. Perineal sensation is preserved, and voluntary anal contraction is normal. Postvoid residual urine volume is 35 mL.

Functional Assessment

At the outpatient assessment, the leg pain NRS score was 8/10 and the low back pain score was 4/10; ODI was 58% (synthetic score, severe functional limitation). Right lower-extremity muscle strength is normal. Left ankle inversion and eversion are also mildly weak, while plantar flexion is normal. Together with lumbar-origin radiating pain and the sensory distribution, these findings support L5 nerve root impairment rather than isolated common peroneal nerve entrapment.

Preliminary Diagnosis

Lumbar disc herniation: left paracentral/lateral recess disc extrusion at L4/5, with mild caudal migration, compressing the traversing left L5 nerve root; left L5 radiculopathy with progressive motor decline. Mild L5/S1 degeneration is an accompanying imaging finding and is not considered the symptomatic level for this episode.

Differential Diagnosis

Common peroneal nerve entrapment: generally does not explain radiating low back and buttock pain, a positive straight-leg raise test, or impaired inversion. Lumbar spinal stenosis: this case is dominated by unilateral radicular symptoms and a single-level soft disc extrusion. Hip disease: no typical groin pain, and hip movement did not provoke the primary symptoms. Infection, tumor, fracture, and cauda equina syndrome: currently lack supporting evidence, but monitoring will continue during hospitalization.

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Surgical Indications and Selection

Radicular pain and functional impairment remain significant after approximately 8 weeks of treatment, and objective, progressive weakness of left ankle dorsiflexion and great toe extension has developed. MRI findings are consistent with localization to the left L5 nerve root. Left-sided L4/5 microdiscectomy and nerve root decompression are planned. There is no spondylolisthesis or definite dynamic instability; fusion or internal fixation is not routinely indicated. The choice of procedure depends on the patient's condition and the surgical team; no claim is made that any particular minimally invasive technique is universally superior to other approaches.

Previous Similar Episodes (Added Teaching Background)

During the two to three years before the current episode, he occasionally experienced aching and a sense of fullness in the lower back after moving many items, bending over to organize goods, or a busy day. This occurred approximately two or three times a year, although he cannot recall the exact frequency. It usually resolved after one or two days of rest, after which he could work as usual. Previously, he had only lower back aching and fullness, without pain extending from the left buttock all the way to the calf and top of the foot as in the current episode, and without foot numbness, difficulty lifting the foot, or tripping while walking. He did not specifically seek medical attention, undergo imaging, or receive prescription treatment for those brief episodes of lower back aching, and he had not been diagnosed with lumbar disc herniation. The similarity to the current episode is lower back discomfort, but the previous episodes did not involve the persistent radiating leg pain and weakness present now. The current episode began with persistent pain after moving boxes on July 9, followed by radiating left leg pain and numbness the next day; difficulty lifting the foot did not develop until September 1. The previous occasional lower back aching must not be counted as part of the duration of this continuous episode.

Daily Life and Context of the Visit (Added Teaching Background)

He is originally from Pidu District, Chengdu, and has worked and lived in Chengdu since marrying. He currently lives in Wuhou District; the detailed address is as recorded in the outpatient registration. He is married and lives with his wife and one daughter. His wife knows about the current lower back and leg discomfort and accompanied him to the visit. He has worked in warehouse management for more than ten years. His usual duties include checking inventory, recording incoming and outgoing stock, and inspecting the warehouse; he sometimes also helps move boxes. His work involves both sitting to make records and standing or walking around, rather than sitting in an office all day. After work, he enjoys watching the news and talking with his family. Before the onset of the lower back and leg symptoms, he would walk around his residential community; he has no regular habit of high-intensity exercise. Recently, the pain has affected his work and sleep. His main concern is that difficulty lifting his foot may affect his walking and work in the future, and he also hopes to understand how treatment can help him recover; this does not mean that he has already agreed to or refused any particular treatment. He came to this hospital because an acquaintance said its orthopaedic care was relatively good, and he also wanted to understand why this episode had not improved. Do not fabricate appointments, referrals, travel time, or prehospital management. In everyday conversation, he may be addressed as Mr. Li. He is not familiar with medical terminology and describes his symptoms in everyday terms such as an aching lower back, pulling pain in the leg, numbness on the top of the foot, and difficulty lifting the foot.

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Neurological Examination and Preoperative Investigations

Muscle Strength Record (MRC Grading)

Muscle group	Left, 8/28	Left, initial outpatient visit	Left, discharge	Right
Hip flexion	5/5	5/5	5/5	5/5
Knee extension	5/5	5/5	5/5	5/5
Ankle dorsiflexion (tibialis anterior)	5/5	3/5 ↓	4/5 ↓	5/5
Great toe extension (extensor hallucis longus)	4/5 ↓	3/5 ↓	4/5 ↓	5/5
Ankle inversion	5/5	4/5 ↓	4/5 ↓	5/5
Ankle eversion	5/5	4/5 ↓	4/5 ↓	5/5
Ankle plantarflexion	5/5	5/5	5/5	5/5

↓ indicates strength below the normal grade of 5/5. Laterality, the responsible spinal level, and nerve root localization are key points for case review.

Preoperative Blood Tests

Samples collected on 2026-09-03; simulated laboratory reference intervals. All results below are within the ranges specified for this case.

Preoperative Blood Test Results

Test	Result	Unit	Reference interval	Flag
White blood cell count	6.8	$\times 10^9$/L	3.5–9.5	
Hemoglobin	151	g/L	130–175	
Platelet count	242	$\times 10^9$/L	125–350	
CRP	1.6	mg/L	<5	
ESR	7	mm/h	0–15	
Creatinine	79	μmol/L	57–97	
ALT	24	U/L	9–50	
Albumin	44	g/L	40–55	
Fasting blood glucose	5.3	mmol/L	3.9–6.1	
Potassium	4.1	mmol/L	3.5–5.3	
Sodium	140	mmol/L	137–147	
INR	1.01		0.8–1.2	
APTT	29	s	25–35	

ECG: sinus rhythm, 78 beats/min, no significant abnormalities. Postvoid residual urine volume: 35 mL. No routine chest CT or electrophysiological testing is specified in this scenario; these may be added if needed based on the clinical differential diagnosis.

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Imaging Reports

Lumbar Spine AP, Lateral, and Flexion–Extension Radiographs

Examination: 2026-08-28 10:00; Report: 2026-08-28 11:00

Findings: Five lumbar vertebral bodies with preserved heights; mild straightening of the physiological lumbar lordosis. Mild narrowing of the L4/5 disc space and mild degeneration at L5/S1. No definite fracture, destructive lesion, or spondylolisthesis. No definite abnormal segmental translation on active flexion–extension views. The examination was completed within the patient's tolerance; pain limited movement, restricting the assessment of dynamic instability.

Impression: Degenerative narrowing of the L4/5 disc space; no definite dynamic instability. Radiographs cannot directly establish nucleus pulposus herniation or nerve root compression.

Purpose of examination: Persistent low back and leg pain during conservative treatment, and preoperative assessment of bony structures and stability. Flexion–extension views are not routinely required for all patients with disc herniation.

Noncontrast Lumbar Spine CT with Coronal and Sagittal Two-Dimensional Reconstructions

Examination: 2026-08-28 10:30; Report: 2026-08-28 12:00

Findings: The designated scan range extends from the inferior margin of T12 to S2, with thin-section noncontrast acquisition and coronal and sagittal reconstructions. A left-sided soft-tissue density is visible along the posterior margin at L4/5, with narrowing of the left lateral recess. No obvious large area of calcified nucleus pulposus. The bony cortices of the vertebral bodies, pedicles, and partes interarticulares are intact. Mild facet degeneration, with no definite pars defect, severe bony spinal canal stenosis, or fracture.

Impression: Possible left-sided posterior disc herniation at L4/5; MRI is recommended to evaluate the nerve roots and soft tissues. No definite bony contraindication to surgery or indication for fixation.

Purpose of examination: Supplementary evaluation of suspected posterior marginal osteophytes/calcification on outpatient imaging and of bony structures. CT does not replace MRI, nor should all patients undergoing surgery for lumbar disc herniation (LDH) undergo repeat CT. The simulated multiplanar reconstruction (MPR) appearance is not a reconstruction from actual volumetric data.

Noncontrast Lumbar Spine MRI

Examination: 2026-09-03 10:10; Report: 2026-09-03 11:05

Findings: The teaching sequence configuration includes sagittal T1-weighted and T2-weighted imaging and axial T2-weighted imaging covering the inferior margin of T12 to S2. The actual MRI scan coverage and axial angulation must be determined according to clinical protocols. The conus medullaris terminates at approximately L1, with no definite signal abnormality. The L4/5 disc shows reduced T2 signal and mild disc space narrowing; a left paracentral/lateral recess nucleus pulposus extrusion with mild caudal migration compresses the left anterior margin of the thecal sac and the traversing left L5 nerve root. The causative lesion is not an extraforaminal herniation at L4/5, nor is compression of the exiting left L4 nerve root the predominant finding. Mild disc bulging at L5/S1, without definite corresponding nerve root compression; no major compressive lesion at the remaining levels. No major signs supporting infection, tumor, or acute fracture.

Impression: Left-sided extruded lumbar disc herniation at L4/5, with caudal migration and left L5 nerve root compression, consistent with the physical signs. The morphology, level, and laterality must be confirmed using the original imaging data; the current images are AI-generated teaching composites and lack diagnostic-grade spatial consistency.

Purpose of examination: MRI was prioritized because of progressive left foot dorsiflexion weakness, without waiting for further nonsurgical treatment. A noncontrast examination is sufficient as the primary imaging configuration for this case.

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Perioperative and Discharge Records

2026-09-03 09:30 | Initial Progress Note

At outpatient assessment, left ankle and great toe dorsiflexion strength was 3/5, decreased from the previous outpatient assessment, consistent with progressive neurological impairment; no cauda equina symptoms. Prioritize MRI and spine surgery assessment, implement fall precautions, suspend symptom-provoking exercises, and document the neurological baseline. Pain improvement alone must not be used to judge neurological safety.

2026-09-03 11:30 | Senior Physician Ward Round

The MRI report was consistent with localization to the left L5 nerve root, establishing a clear indication for surgery. The benefits and risks of continued nonsurgical treatment versus early decompression were compared, and surgery as soon as possible was chosen. Repeat examination showed no urinary retention or altered perineal sensation; left dorsiflexion strength remained 3/5 without further decline.

2026-09-03 16:00 | Preoperative Discussion

Planned left L4/5 microdiscectomy and decompression of the left L5 nerve root. No definite instability, spondylolisthesis, or predominant mechanical low back pain; fusion is not routinely indicated. Discussed risks of dural tear/cerebrospinal fluid leak, nerve injury, infection, hematoma, recurrence, residual numbness or incomplete strength recovery, wrong-level surgery, and reoperation. General anesthesia assessment: ASA I; simulated informed consent, with no actual signature generated.

2026-09-04 08:30–09:45 | Operative Record

Following general anesthesia, the patient was positioned prone with the abdomen free of pressure. Fluoroscopy localized L4/5, and the level and left side were verified. Cefazolin 2 g was administered intravenously for prophylaxis 30 minutes before skin incision. A short left-sided incision and limited paraspinal muscle dissection were performed, followed by a limited interlaminar fenestration under the microscope, preserving sufficient facet joint structures. The left L5 nerve root was exposed, revealing ventral compression by extruded nucleus pulposus. Extruded and loose nucleus pulposus material was removed, avoiding excessive disc curettage. Exploration confirmed release of the nerve root and restoration of dural sac pulsation. Hemostasis was achieved, with no cerebrospinal fluid leak observed; no internal fixation or fusion was performed. Estimated blood loss was 30 mL. Surgery proceeded smoothly, with no transfusion. Synthetic pathology conclusion for the submitted specimen: degenerative fibrocartilage/nucleus pulposus tissue.

2026-09-04 13:00 | Initial Postoperative Progress Note

Awake, with stable vital signs. Left leg pain decreased to 2/10; incisional pain was 3/10. Left ankle dorsiflexion strength was 3/5 and great toe dorsiflexion strength was 3/5. Reduced sensation persisted; toes were warm. No new perineal numbness or difficulty urinating, and no significant incisional bleeding. Short-term analgesia, neurological examinations, mechanical prophylaxis, and early mobilization; no routine repeat CT/MRI in the absence of indications arising from abnormal findings.

2026-09-05 09:00 | Postoperative Day 1

Left leg pain was 1–2/10. Left ankle and great toe dorsiflexion strength improved to 4/5, and numbness over the dorsum of the foot decreased. The wound was dry, and spontaneous urination was normal. Mobilize out of bed and walk with assistance; use a short-term ankle-foot orthosis if needed to prevent tripping. Prophylactic antibiotics were stopped within 24 hours after surgery; reassess venous thrombosis prevention and management according to mobility, bleeding, and individual risk.

2026-09-06 09:00 | Postoperative Day 2

No fever, new leg pain, progressive weakness, or bowel or bladder abnormalities. Walking tolerance improved; left dorsiflexion strength was 4/5, and reflexes were symmetrical bilaterally. Continue staged rehabilitation, wound monitoring, and smoking cessation education; do not promise immediate full recovery of muscle strength.

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2026-09-07 09:00 | Discharge Summary

Discharge diagnoses: Left-sided extruded lumbar disc herniation at L4/5 with left L5 radiculopathy; status post left-sided L4/5 microdiscectomy and nerve root decompression. The wound is dry; leg pain is 1/10; left ankle dorsiflexion strength is 4/5 and great toe dorsiflexion strength is 4/5; mild numbness over the dorsum of the foot persists. Able to walk with protective assistance; bladder and bowel function are normal. Outcome: Improved. No long-term antibiotics or routine prophylactic anticoagulation prescription is required; reassess based on individual risk.

Discharge Medication and Follow-up

Acetaminophen 500 mg orally as needed, at intervals of at least 6 hours; for this case, a maximum of 2 g/day for no more than 3 days. Avoid duplicate use with other preparations containing the same ingredient. Recheck the incision and neurological function 10–14 days after surgery, with a rehabilitation follow-up at 4–6 weeks. Gradually increase walking; avoid repeated bending, twisting, and heavy lifting during early recovery. Assess return to work according to recovery. Seek emergency care immediately for new or worsening muscle weakness, urinary retention, perineal numbness, or abnormal bowel function; seek prompt medical attention for fever or incision redness, swelling, or drainage.

Review Status

The entire clinical course, all doses, and all examinations and investigations are teaching constructs and have not yet been signed off and confirmed by a clinical expert. There is no follow-up imaging or actual evidence of muscle strength recovery. This case is intended for discussion of surgical indications, localization, choice of surgical technique, and perioperative care; it does not claim a guaranteed treatment outcome.

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AP Lumbar Spine View · Synthetic Image

Patient information

Patient	Sex	Age	Case ID
Li Zhiqiang (fictional)	Male	46 years	SIM-LDH-20260903

AP view; single-projection example

AI-generated synthetic image, not acquired by imaging equipment or a real DICOM image. The different planes were not reconstructed from the same volumetric dataset; laterality, continuity between slices, and tissue morphology may contain discrepancies. Pending review by an imaging expert.

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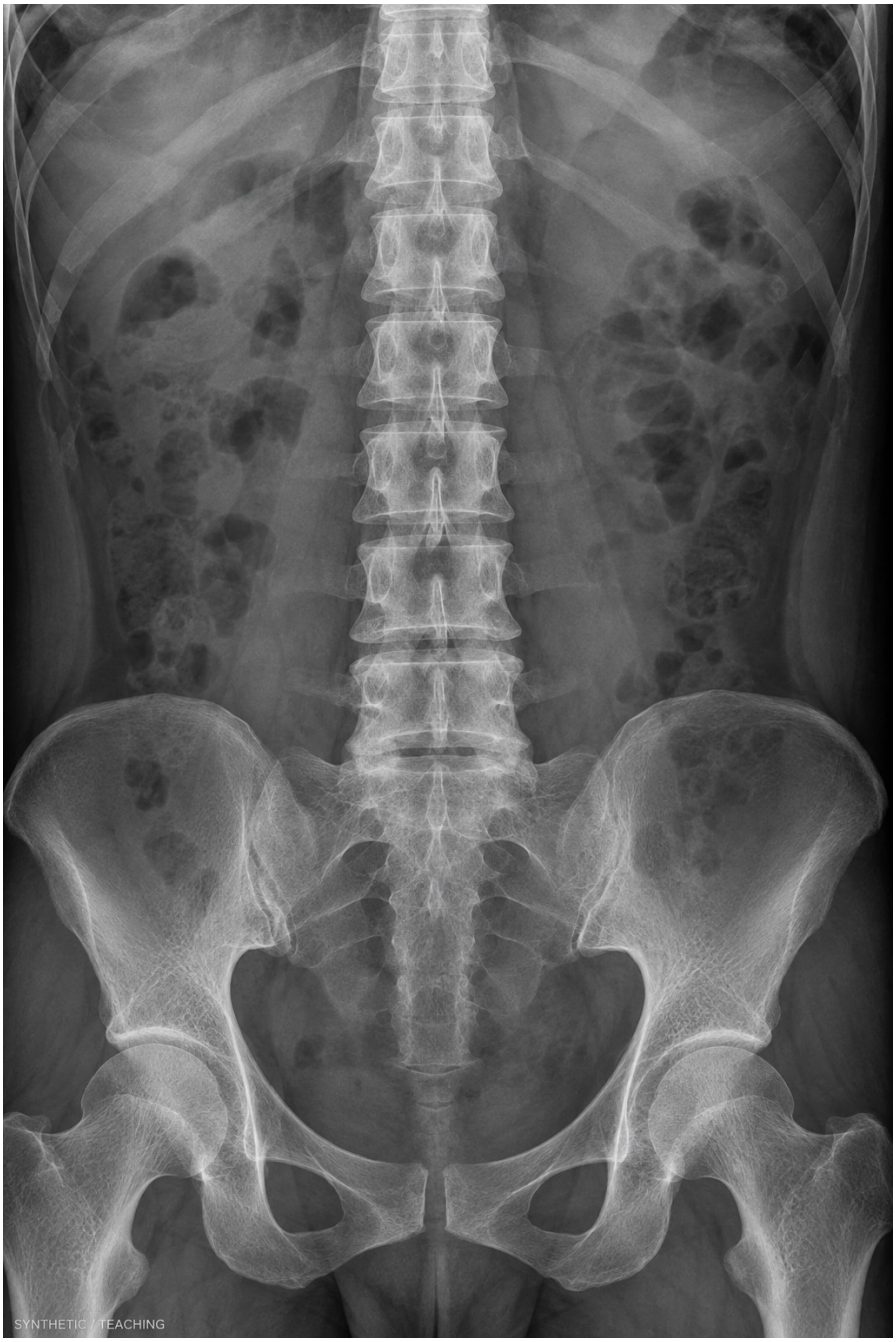

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Lateral Lumbar Spine View · Synthetic Image

Patient: Li Zhiqiang (fictional), male / 46 years old, SIM-LDH-20260903

Lateral view; single-projection example

[Unacceptable image quality; interface demonstration only] The generated image contains too many lumbar-like vertebral bodies, inconsistent with the case specification of five lumbar vertebrae. For interface demonstration only; must not be used for vertebral level localization.

AI-generated synthetic image, not acquired by imaging equipment or a real DICOM image. The different planes were not reconstructed from the same volumetric dataset; laterality, continuity between slices, and tissue morphology may contain discrepancies. Pending review by an imaging expert.

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SOURCE SECTION 12

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Case Record Compilation in Photocopy Format · Not a Real Medical Document

Flexion Functional View · Synthetic Image

Patient: Li Zhiqiang (fictional) | Male / 46 years old | SIM-LDH-20260903

Lateral flexion view; single-projection example

[Unacceptable Image Quality; Interface Demonstration Only] The generated image contains too many lumbar-like vertebral bodies. For interface demonstration only; it cannot be used to calculate actual range of motion or assess instability.

AI-generated image, not acquired by imaging equipment or a real DICOM image. The different planes are not reconstructed from the same volumetric dataset; laterality, interslice continuity, and tissue morphology may be inaccurate. Pending review by an imaging specialist.

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Simulated Teaching Hospital / Spine Surgery

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Extension Functional View · Synthetic Image

Patient: Li Zhiqiang (fictional) Male / 46 years old SIM-LDH-20260903

Lateral extension view; single-exposure example

[Unacceptable image quality; interface demonstration only] The generated image contains too many lumbar-like vertebrae. For interface demonstration only; it cannot be used to calculate actual range of motion or assess instability.

AI-generated synthetic image, not acquired by imaging equipment or a real DICOM image. The different planes were not reconstructed from the same volumetric dataset; laterality, continuity between slices, and tissue morphology may be inaccurate. Pending review by an imaging expert.

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Simulated Teaching Hospital / Spine Surgery

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Axial CT · Synthetic Images

Patient: Li Zhiqiang (fictional) · Male / 46 years old · SIM-LDH-20260903

Axial images · T12–S2; 16 representative slices, arranged from left to right and top to bottom.

AI-generated synthetic images, not acquired by imaging equipment and not real DICOM images. The planes were not reconstructed from the same volumetric dataset; laterality, continuity between slices, and tissue morphology may contain discrepancies. Pending review by an imaging specialist.

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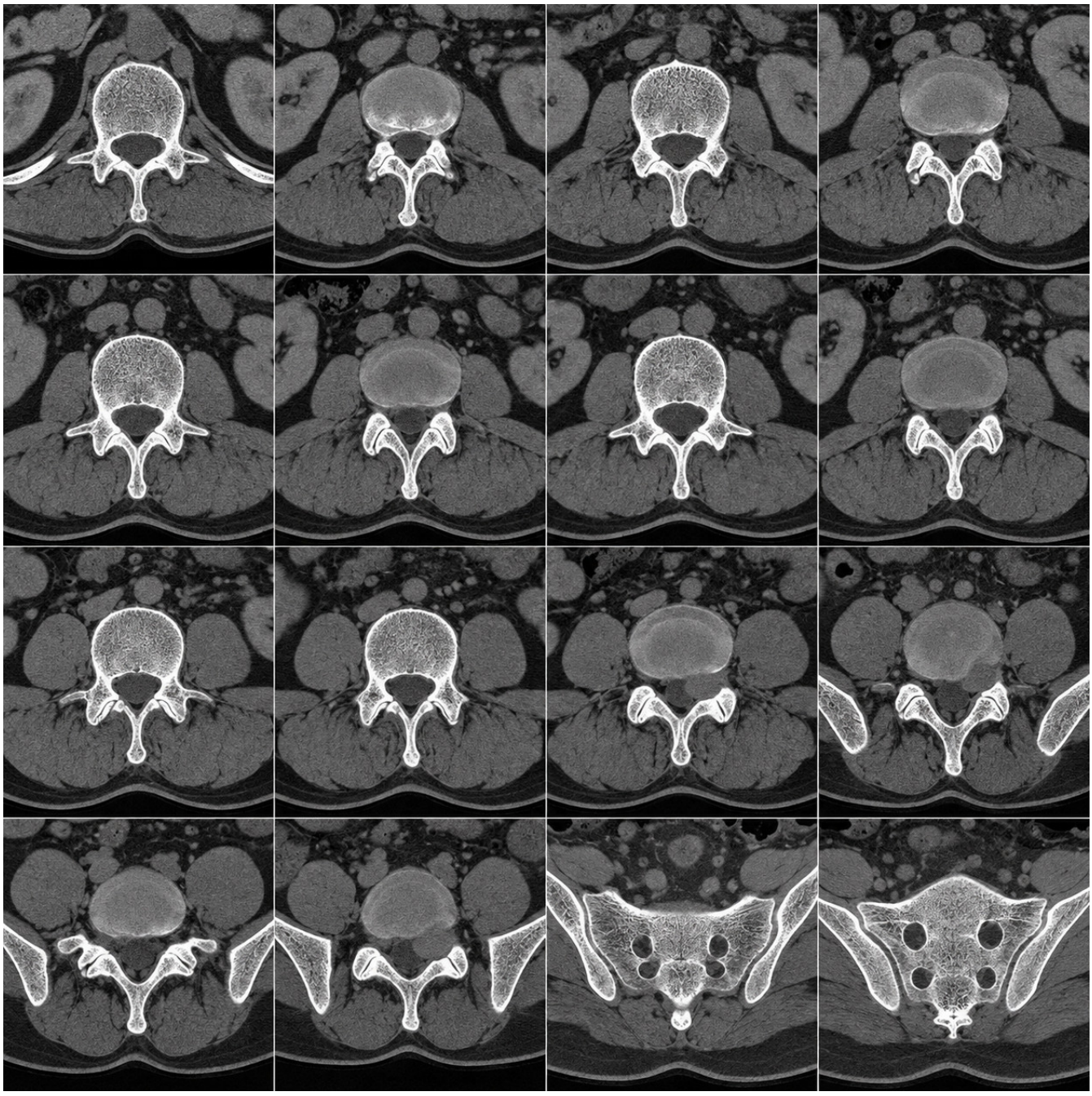

SOURCE SECTION 15

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CT Coronal Reconstructions · Synthetic Images

Patient: Li Zhiqiang (fictional) · Male / 46 years old · SIM-LDH-20260903

Coronal view · Simulated MPR; 4 representative slices, arranged from left to right and top to bottom

AI-generated images, not acquired by imaging equipment or real DICOM images. The planes were not reconstructed from the same volumetric dataset; laterality, continuity between slices, and tissue morphology may contain discrepancies. Pending review by an imaging expert.

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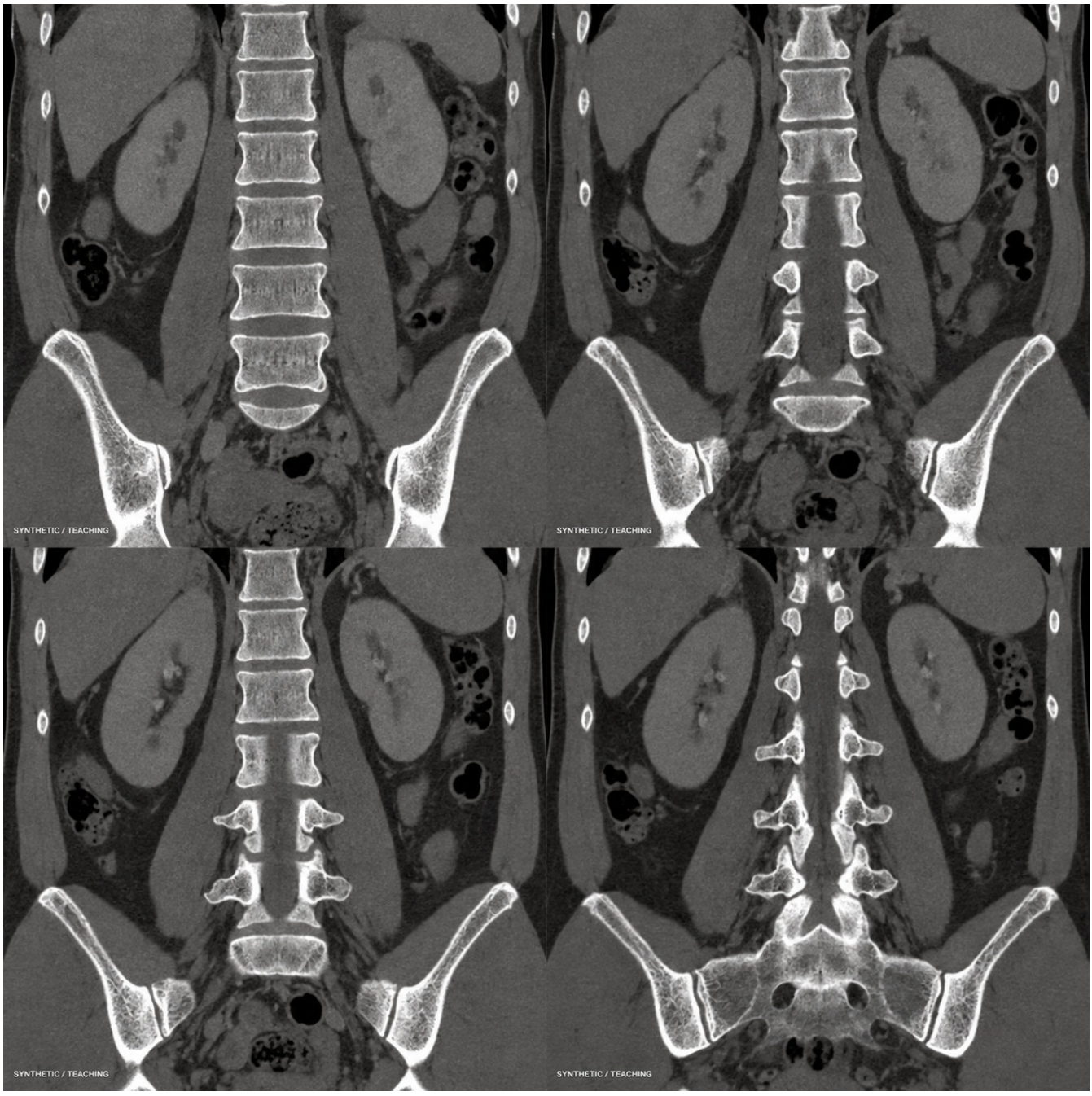

SOURCE SECTION 16

Simulated Teaching Hospital / Spine Surgery

Photocopy-style case compilation · Not a real medical document

CT Sagittal Reconstructions · Synthetic Images

Patient: Li Zhiqiang (fictional) | Male / 46 years old | SIM-LDH-20260903

Sagittal views · Simulated MPR; 4 representative slices, arranged from left to right and top to bottom.

AI-generated synthetic images, not acquired by imaging equipment or derived from real DICOM data. The planes are not reconstructed from the same volumetric dataset; laterality, interslice continuity, and tissue morphology may contain discrepancies. Pending review by an imaging expert.

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SOURCE SECTION 17

Simulated Teaching Hospital / Spine Surgery

Photocopy-style case compilation · Not a real medical document

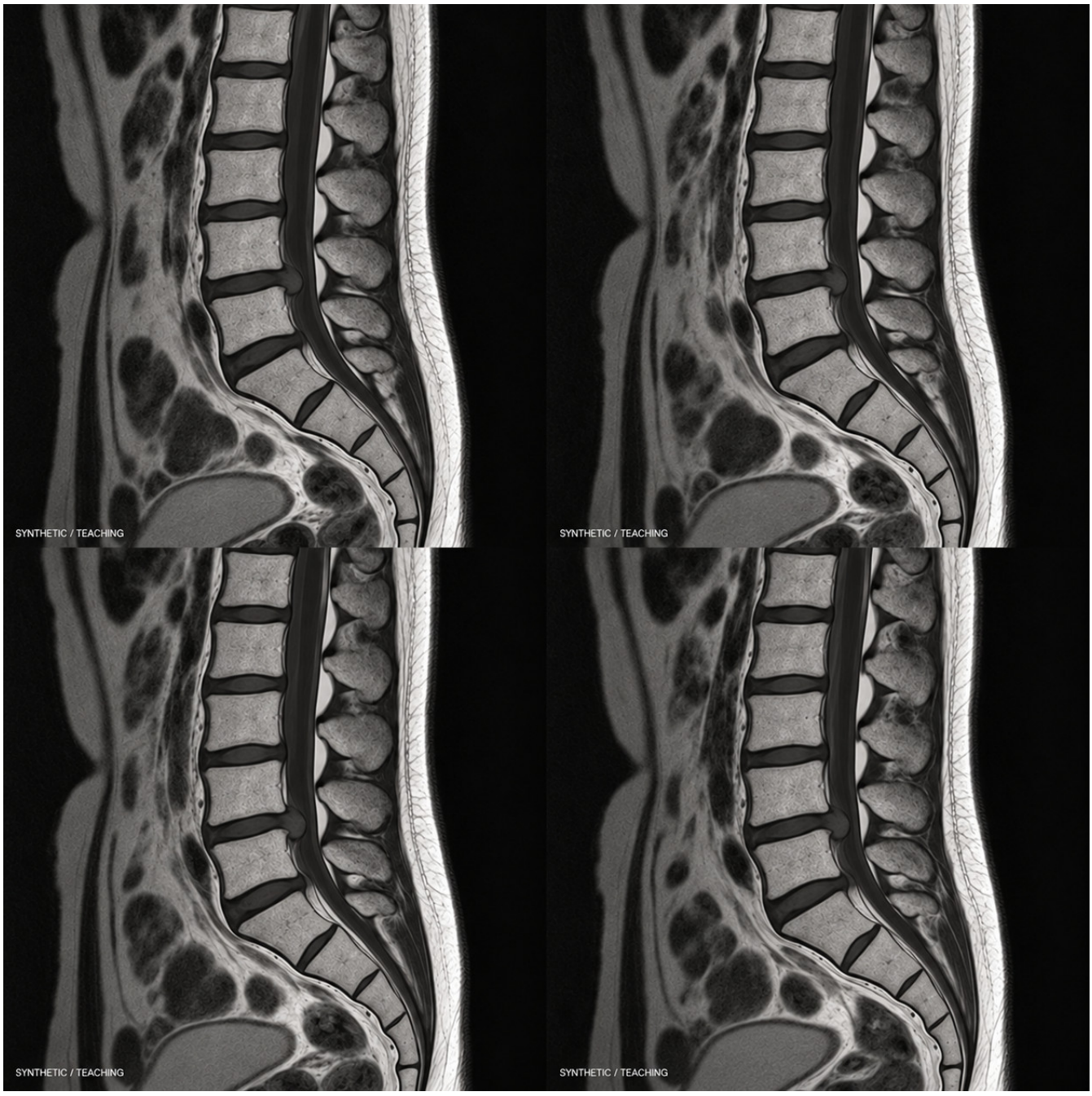
MRI T1WI, Sagittal View · Synthetic Images

Patient: Li Zhiqiang (fictional) · Male / 46 years old · SIM-LDH-20260903

T1WI · Sagittal view; 4 representative slices, arranged from left to right and top to bottom.

AI-generated synthetic images, not acquired by imaging equipment or real DICOM images. The planes were not reconstructed from the same volumetric dataset; laterality, continuity between slices, and tissue morphology may contain discrepancies. Pending review by an imaging expert.

SIM-LDH-20260903 | Synthetic teaching case · Pending expert review · v1.1 · Materials synchronized on 2026-09-14



SOURCE SECTION 18

Simulated Teaching Hospital / Spine Surgery

Photocopy-style case compilation · Not a real medical document

MRI T2WI, Sagittal View · Synthetic Images

Patient: Li Zhiqiang (fictional), male / 46 years old, SIM-LDH-20260903

T2WI · Sagittal view; 4 representative slices, arranged from left to right and top to bottom.

AI-generated synthetic images, not scanner-acquired images or real DICOM data. The planes were not reconstructed from the same volumetric dataset; laterality, continuity between slices, and tissue morphology may contain discrepancies. Pending review by an imaging expert.

SIM-LDH-20260903 | Synthetic teaching case · Pending expert review · v1.1 · Data synchronized on 2026-09-14

SOURCE SECTION 19

Simulation Teaching Hospital / Spine Surgery

Case Record Compilation in Photocopy Format · Not a Real Medical Document

MRI T2WI, Axial · Synthetic Images

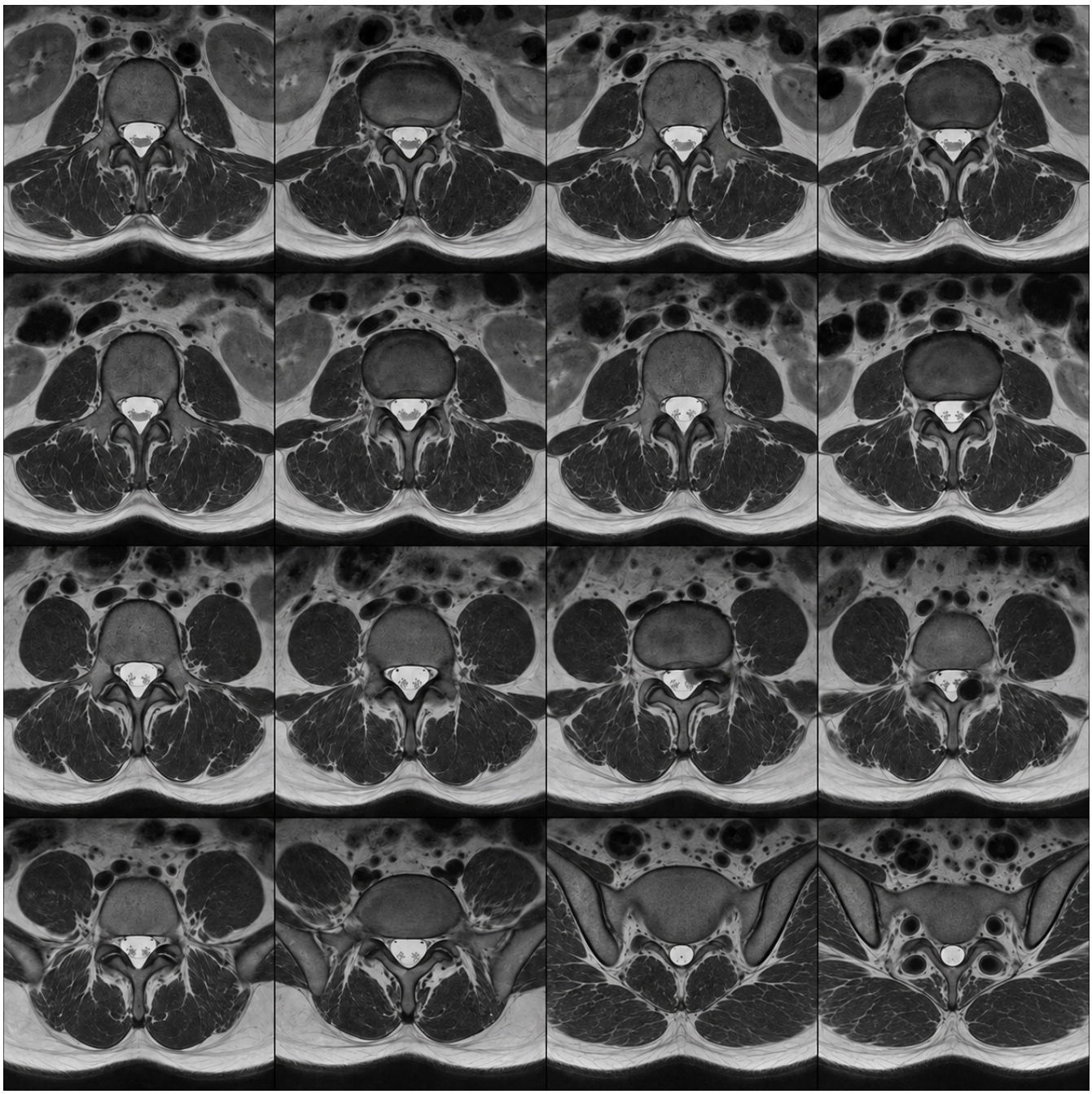
Patient Information

Patient	Sex	Age	Case ID
Li Zhiqiang (fictional)	Male	46 years	SIM-LDH-20260903

T2WI · Axial · T12–S2; 16 representative slices, arranged from left to right and top to bottom.

AI-generated synthetic images, not scanner-acquired images or real DICOM data. The planes were not reconstructed from the same volumetric dataset; laterality, interslice continuity, and tissue morphology may contain discrepancies. Pending review by an imaging expert.

SIM-LDH-20260903 | Synthetic Teaching Case · Pending Expert Review · v1.1 · Data synchronized on 2026-09-14



Simulated Teaching Hospital / Spine Surgery

Photocopy-style case compilation · Not a real medical document

Expert Review and References

Please review the clinical scenario and image quality separately; this edition has not passed medical review. Descriptions of imaging abnormalities are targets for the synthetic case and do not mean that the generated pixels have been confirmed to support the diagnosis.

Expert review

Review item	Expert opinion
History, nerve root localization, and changes in muscle strength	
Surgical indications, timing, and procedure selection	
Vertebral counting on X-rays and consistency across functional-position views	
CT/MRI levels, laterality, and morphology of nerve root compression	
Whether each sequence is suitable for teaching or requires replacement	
Perioperative, discharge, and rehabilitation arrangements	

Conclusion: Suitable / Suitable after revision / Unsuitable

Expert: _____ Date: _____

References

WFNS: Surgical Recommendations for Lumbar Disc Herniation (published in 2024)

Supports individualized consideration of surgery in circumstances such as failed nonsurgical treatment or progressive neurological impairment; fusion is not routinely recommended for primary uncomplicated disc herniation.

WFNS Spine Committee: Diagnostic and Treatment Recommendations

Establish clinical localization using the history, muscle strength, sensation, sphincter function, and straight leg raise test; MRI is the principal examination for confirming herniation and soft-tissue compression.

ACR Appropriateness Criteria: Low Back Pain

Appropriate imaging should be selected for patients with persistent or progressive symptoms who are being considered for surgery; this case involves declining motor function and requires timely evaluation.

Reference consultation and compilation date: 2026-09-10. Complete editable case data and generation prompts are retained with the project.

SIM-LDH-20260903 | Synthetic teaching case · Pending expert review · v1.1 · Materials synchronized on 2026-09-14